Nicholas Amalraj

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Education

University of Michigan — Ann Arbor, Michigan
B.S.E. in Computer Science, Minor in Data Science — *May 2027*
Relevant Coursework: *Data Structures and Algorithms, Web Systems, Computer Networks, Operating Systems*

Experience

Software Engineering Intern — May 2026 – Present
Consumers Energy — *Jackson, MI*

- Built a full-stack operations platform with React, TypeScript, .NET, and SQL, to bundle geographically proximate work orders, lowering redundant dispatches by **20%** and improving resource utilization
- Eliminated conflicting writes across multi-user workflows by building an event-driven locking service with Kafka, using lock/unlock event streams to coordinate concurrent bundle edits and preserve data consistency
- Instrumented OpenTelemetry traces and metrics across backend APIs and user workflows, integrating with Grafana to track user activity and database latency, enabling data-driven analysis and faster diagnosis of production issues

Software Engineering Intern — May 2025 – August 2025
Consumers Energy — *Jackson, MI*

- Designed a web app with .NET, HTML/CSS, SQL, REST APIs, and Azure to streamline asset health data collected by drones, allowing field operators to monitor **8M+** electric grid components and onboard legacy assets
- Engineered an export system to transfer asset data into a machine learning pipeline, reducing manual entry by **95%** and automating predictive maintenance, saving **$500k annually**
- Shipped production features end-to-end using TDD and automated CI/CD pipelines, leveraging blue-green deployments to enable zero-downtime releases

Machine Learning Intern — May 2024 – August 2024
Michigan AI Lab — *Ann Arbor, MI*

- Engineered a PyTorch transformer for drought prediction, accelerating training by **13×** over existing climate forecasting models while achieving a **0.91 F2 score** on a held-out test set
- Benchmarked state-of-the-art climate forecasting models on 20 years of environmental data (**22M+ records**), establishing performance baselines for transformer evaluation
- Initiated a custom Explainable AI framework project in SHAP, enabling the visualization of thousands of model decisions to foster transparency and trust for users

Projects

Search Engine | *Python, Flask, SQLite, HTML/CSS, AWS*

- Architected an inverted index pipeline using MapReduce to compute TF-IDF scores across **3,000+** documents, maximizing search options and allowing for more variety in results
- Developed an index server to calculate cosine similarity between documents and queries, incorporating PageRank scores into a weighted ranking system for relevant results
- Deployed the search engine to AWS with EC2, Nginx, and Gunicorn, ensuring reliable real-time search capabilities

World Cup RAG Pipeline (Live Demo) | *Python, ChromaDB, SentenceTransformers, Gemini 2.5 Flash, Gradio*

- Architected a Retrieval-Augmented Generation (RAG) system using ChromaDB and all-MiniLM-L6-v2 embeddings, achieving an **86% Hit@5 retrieval rate** across 1,260 semantic chunks
- Integrated the pipeline to make a grounded question-answering application with Gemini 2.5 Flash LLM, source attribution, retrieval diagnostics, and a Gradio frontend to deliver explainable, citation-backed responses

Virtual Memory Pager | *C++*

- Built a demand-paged multi-process virtual memory pager in C++, implementing copy-on-write forking, swap and file-backed pages, and MMU integration
- Implemented a clock-based page replacement policy and caching strategy, reducing disk I/O by **20%** under memory-constrained workloads

Skills

Languages: Python, C++, C#/.NET, JavaScript, TypeScript, HTML/CSS, SQL
Frameworks: React, ASP.NET, Flask, Node.js
Tools: Git, AWS, Azure, Hadoop, Agile, Linux, Kafka, OpenTelemetry (OTel), Grafana
Libraries: pandas, NumPy, Matplotlib, Scikit-learn, TensorFlow, PyTorch, Boost